Start

About me

Researcher interested in the intersection of Physics and AI. Data-Driven Physics Discovery and Physics-Inspired ML theory

Areas of specialization

• Physics • Machine Learning • Geophysics • Seismic Inversion • High Energy Physics

Programming

- python
- C,C++
- Matlab
- Fortran
- html, css, php

Other Skills

• Proficient Arabic Speaker
• Freelance Photographer

Interests

Physics / AI / Photography
Physics / AI / Photography
Physics / AI / Photography

@ ibrahime.org
in ibrahimelsharkawy

Work Experience

March 2025 - **Doctoral Researcher**
Lawrence Berkeley National Laboratory · Berkeley, CA
• Researcher at the National Energy Research Scientific Computing Center (NERSC) **building Foundation Models for Scientific Point Clouds/Graphs** for High Energy Physics, Cosmology and Molecular Dynamics.

2024-2025 **Research Geophysicist Intern**
ExxonMobil · Houston, TX
• Researched and developed pioneering machine learning **parameter estimation methods for 3D-seismic inversion** with the potential of enabling the inversion of traditionally degenerate earth properties • Researched and developed **novel ML anomaly detection tools** for 4D Seismic.

2018-2023 **Research Geophysics and Software Engineering Intern**
Petroleum GeoServices · Houston, TX
• Researched **ML methods for first break picking** in seismic. • Built a seismic ML anomaly detector • Worked on **cloud migration** and microservices

Education

2025- **Physics**
PhD · University of Toronto

2023-2025 **Physics**
MSc · University of Illinois Urbana-Champaign

2019-2023 **Computational Physics**
B.S. · Rice

2019-2023 **Applied Mathematics and Philosophy**
B.A. · Rice

Awards and Honors

2025 University of Toronto Connaught International Fellow
2024 1st Place Higgs Uncertainty Challenge @ NeurIPS, CERN
2023 3-Year UIUC Graduate College Fellowship
2021 J & M Graham Physics Scholar
2019 Shell Eco-Marathon
2016 Eagle Scout

Research Experience

2025- **Foundation Models For Science**
LBNL· Prof. Wahid Bhimji, Prof. Ben Nachman · Berkeley, CA
• Building foundation models for scientific point clouds with **cross-domain transfer.**• Researching **optimal pre-training objectives** for foundation models in science, where simulators can produce *labeled* data at scale.

2023-2025 **Neural Uncertainty Quantification and Parameter Estimation**
UIUC · Prof. Yoni Khan Prof. Ben Hooberman · Urbana, IL
• Used an effective field theory of NN to **successfully predict** initialization-driven ensemble uncertainty without training [1].• Built a **competition-winning** ML methodology robust to systematic sources of error [2]

2019-2023 **CERN CMS Higgs-W-W-Boson Analysis**
Rice **Physics** · Prof. Karl Ecklund · Houston, TX
• Working on the CMS Same-Flavor W^+W^- Boson Pair Cross Section Measurement in Proton-Proton Collisions at $\sqrt{s} = 13$TeV.

Select Papers

1. Elsharkawy, Ibrahim, et al. *"Uncertainty Quantification from Ensemble Variance Scaling Laws in Deep Neural Networks."* Machine Learning: Science and Technology, vol. 6, no. 3, 26 Aug. 2025, p. 035040
2. Elsharkawy, Ibrahim, and Yonatan Kahn. *"Contrastive Normalizing Flows for Uncertainty-Aware Parameter Estimation."* arxiv.org/abs/2505.08709.
3. Elsharkawy, Ibrahim, et al. *"OmniMol: Transferring Particle Physics Knowledge to Molecular Dynamics with Point-Edge Transformers."* ArXiv.org, 2026, arxiv.org/abs/2601.10791.
4. Wahid Bhimji, et al. "FAIR Universe HiggsML Uncertainty Dataset and Competition." , In Proceedings, NeurIPS 2025 Dataset and Competition Track
5. Mikuni, Vinicius, et al. *"OmniCosmos: Transferring Particle Physics Knowledge across the Cosmos."* ArXiv.org, 2025, arxiv.org/abs/2512.24422.

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Select Talks/Posters

1. (Invited Talk) *1st Place Competition Milestone (Ensembles and Uncertainty Quantification)* in The Challenge of Handling Uncertainties in Fundamental Science @ NeurIPS 2024
2. (Invited Talk) *Contrastive Normalizing Flows for Uncertainty Aware Parameter Estimation* @ CERN 7th Inter-Experimental LHC Machine Learning Workshop 2025
3. (Poster) Wahid Bhimji, et al. *"FAIR Universe HiggsML Uncertainty Dataset and Competition."*, NeurIPS 2025 Dataset and Competition Track
4. (Poster) I. Elsharkawy, et.al, *Uncertainty Quantification from Scaling Laws in Deep Neural Networks*, in Machine Learning and the Physical Sciences Workshop @ NeurIPS 2024
5.(Poster) I. Elsharkawy, et.al,*Pre-Training For Science: A study on Foundation Model Training Objectives*, Stanford Center of Decoding the Universe Forum 2025

Teaching Experience

1. TA and Lecturer *University of Toronto, 2026, Physics 2108/2109, "The Physics of Machine Learning"*
2. TA *University of Toronto, 2025, Physics 252, "Thermal Physics"*
3. TA *University of Illinois Urbana-Champaign, 2024, Physics 413 "Optics Lab"*